Super-Resolution in Waveform-Limited OFDM ISAC: Capability and Robustness of Staged MUSIC

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Abstract

This project studies the capability and robustness of MUSIC-based super-resolution in waveform-limited OFDM integrated sensing and communications (ISAC) environments. Motivated by earlier work showing clear MUSIC upside in a 1-D range-only study, this project asks whether that advantage survives into a 3-D staged pipeline operating on communications-limited waveforms with fragmented occupancy, structured clutter, and coherent multipath.

The project compares a staged masked MUSIC estimator (with spatial FBSS and support-aware range/Doppler FBSS) against a fair masked FFT baseline with local refinement, both operating on the same de-embedded OFDM observation. The evidence consists of a saved FR1 Monte Carlo bundle with 64 trials per sweep point across three indoor industrial scene classes and nine public sweep families, plus saved 64-trial follow-on analyses for expected-order MUSIC and controlled nuisance-strength sweeps.

The main findings are: (1) under the default MDL model-order path, MUSIC wins strongly only in the intersection scene, while open aisle favors the FFT baseline and rack aisle collapses; (2) model-order estimation via MDL is a real bottleneck, because expected-order MUSIC recovers open-aisle performance to parity with FFT and strengthens the already-clear intersection win; (3) in clutter-dominated regimes such as rack aisle, fixing the model order does not help, and the saved stage diagnostics show nuisance-aligned azimuth candidates capturing the search before later stages can recover; (4) the central contribution is therefore a resolution-versus-robustness map rather than a generic superiority claim.

The project thus positions itself as a regime-boundary study of where and why MUSIC super-resolution helps in realistic OFDM ISAC, identifying the roles of model-order control and clutter sensitivity in determining whether the earlier 1-D upside survives in 3-D staged operation.

1 Introduction and Motivation

Integrated sensing and communications is a natural 6G research direction because future infrastructure is expected to share spectrum, aperture, computation, and deployment footprint across both links and sensing tasks [1]. A base station that already transmits a wideband MIMO-OFDM waveform is an obvious candidate sensor. If the communication waveform can support useful range, Doppler, and angle estimation, then the network gains environmental awareness without a separate radar band or separate radar front end.

The challenge is that a communications waveform is not a radar waveform. In a scheduled OFDM grid, only a subset of resource elements may be occupied, pilots may be sparse, data symbols may be unknown to the sensing receiver, and the occupied support may be fragmented across frequency and slow time. These constraints weaken the three ingredients that a classical super-resolution pipeline wants:

- contiguous bandwidth for fine range structure,
- contiguous slow-time aperture for Doppler structure,
- and covariance matrices whose dominant subspaces remain stable in the presence of missing support and coherent nuisance paths.

The project question is therefore not "does MUSIC work on an ideal radar cube?" It is:

> Can MUSIC, aided by FBSS, provide practically useful super-resolution gains over a fair FFT baseline when sensing is constrained by a communications-style OFDM resource grid?

That question is relevant to 6G because it tests whether subspace methods survive realistic waveform limitations rather than only idealized radar assumptions.

2 Problem Statement and Study Scope

The project focuses on a narrow but important scope:

- two-target indoor industrial scenes,
- monostatic OFDM sensing from a private-5G-style FR1 waveform anchor,
- communications-limited resource grids with fragmented occupancy,
- a fair comparison between a masked FFT baseline and a masked staged MUSIC estimator,
- and a submission-grade Monte Carlo bundle saved under `results/submission`.

The report answers four concrete questions:

1. What does waveform-limited OFDM sensing do to the super-resolution problem?
2. How are MUSIC and FBSS adapted to this masked observation setting?
3. What does the saved submission bundle actually show across nominal points and sweeps?
4. Which conclusions are robust, and which require careful framing?

The report does **not** claim a true joint 3-D MUSIC implementation, an unknown-data receiver, a multi-target tracking system, or a hardware demonstration. It is a simulation study of estimator behavior under waveform and scene constraints.

2.1 Relation to Prior Work

This project sits at the intersection of two established research lines. First, recent ISAC literature frames sensing-from-communications-waveforms as a central 6G theme, with OFDM and MIMO-OFDM appearing as natural candidates because they already dominate practical wireless systems [1]. Earlier joint radar-communications work also showed why OFDM is attractive for dual use: the same multicarrier waveform can support communication while enabling delay- and Doppler-sensitive sensing processing on the recovered symbols [2, 3].

Second, the super-resolution machinery used here comes from classical array processing rather than from new ISAC-specific theory. MUSIC is the canonical subspace method for super-resolution parameter estimation [4], while spatial smoothing and forward-backward spatial smoothing are standard remedies for source coherence and multipath-induced covariance rank loss [5, 6].

2.2 Motivating 1-D Result

The starting point for this project is a clean 1-D range-only comparison. Figure 1 shows MUSIC versus FFT for two targets separated by 0.70 resolution cells (sub-Rayleigh) with 96 subcarriers, 16 snapshots, and 20 dB SNR under full support.

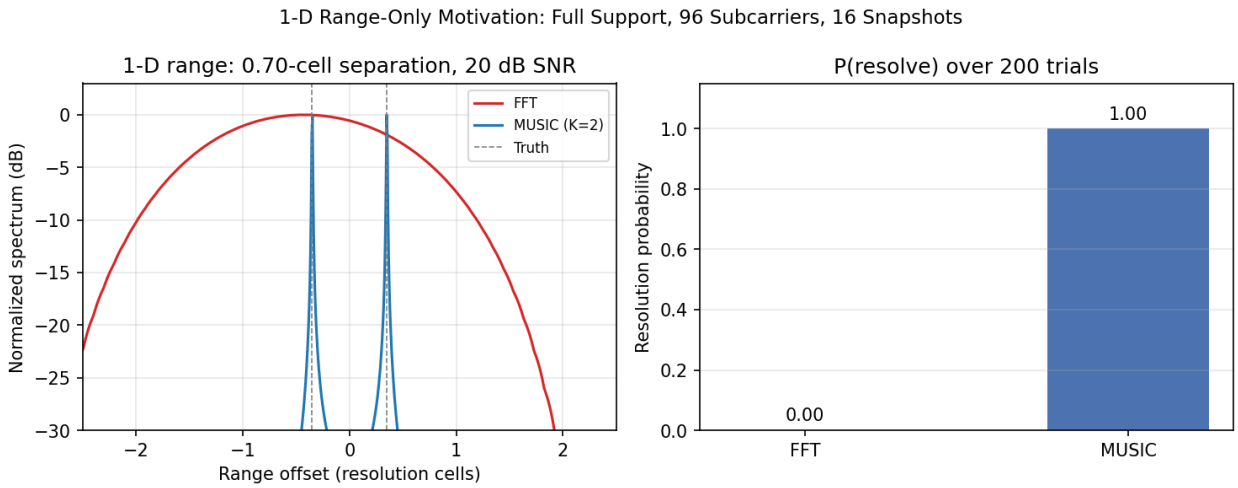

Figure 1: 1-D range-only motivation. With full frequency support and known model order ($K = 2$), MUSIC resolves both targets at 0.70-cell separation in 100% of 200 trials, while FFT cannot separate them at all. This project asks how much of this advantage survives when the observation is masked by communications scheduling, the model order is unknown, and structured clutter is present.

In this clean setting, MUSIC's super-resolution advantage is unambiguous: perfect resolution at sub-Rayleigh separation where FFT produces a single merged peak. The project question is what happens to this advantage under waveform limitation.

The novelty claim of this report is therefore intentionally narrow. It is not proposing a new waveform, a new FBSS theorem, or a new joint estimator. Instead, it asks whether a staged, support-aware masked MUSIC pipeline still provides value once sensing is forced onto a communications-style OFDM resource grid with fragmented support and structured nuisance multipath. Limited-symbol-knowledge sensing remains an important future extension, but the final claim set here is anchored to known-symbol sensing so that the super-resolution mechanisms are not conflated with receiver-side symbol-recovery failure.

3 Signal Model and Mathematical Background

3.1 Waveform-Limited OFDM Observation Model

Let h index the virtual horizontal channel, k the subcarrier, and n the slot-spaced slow-time snapshot index. The masked communications-limited observation can be written schematically as

$$y_{h,k,n} = m_{k,n} x_{k,n} \left(\sum_{p=1}^{P} \alpha_p a_h(\theta_p) e^{-j2\pi f_k \tau_p} e^{j2\pi \nu_p t_n} s_p[n] \right) + w_{h,k,n}. \tag{1}$$

where

- $m_{k,n} \in \{0, 1\}$ is the resource-support mask,
- $x_{k,n}$ is the transmitted OFDM symbol on that resource element,
- α_p is the path gain,
- θ_p, τ_p, and ν_p are azimuth, delay, and Doppler,
- $a_h(\theta_p)$ is the spatial steering term,
- $s_p[n]$ is the slow-time source process for path p, sampled once per slow-time snapshot,
- and $w_{h,k,n}$ is complex receiver noise.

For a radar interpretation, delay and Doppler are converted to range and velocity by

$$R = \frac{c\tau}{2}, \qquad v = \frac{\lambda\nu}{2}. \tag{2}$$

If the sensing receiver knows the transmitted symbols on a subset of occupied resource elements, it can de-embed them and form a masked sensing cube

$$\tilde{y}_{h,k,n} = \begin{cases} y_{h,k,n}/x_{k,n}, & (k,n) \in \Omega_{\text{known}}, \\ 0, & \text{otherwise}. \end{cases} \tag{3}$$

This is the core object used by both headline estimators in the project. Unlike a radar-dedicated cube, it contains structured zeros and irregular support inherited from communications scheduling and symbol knowledge.

3.2 Why Communications Scheduling Makes Super-Resolution Hard

Three OFDM limitations matter immediately:

1. **Fragmented frequency support.** Missing subcarriers reduce effective bandwidth and disrupt the covariance structure used by both FFT and subspace methods.
2. **Fragmented slow-time support.** Missing slow-time snapshots (here sampled once per slot) reduce coherent processing interval and Doppler stability.
3. **Limited transmitter knowledge.** If only pilot symbols are known, then the sensing receiver cannot cleanly de-embed data-bearing resource elements without a more complete communication receiver.

These are exactly the limitations that make the problem relevant to 6G ISAC rather than classical radar.

3.3 MUSIC and FBSS in This Setting

MUSIC works by decomposing a covariance matrix into signal and noise subspaces and evaluating a pseudospectrum of the form [4]

$$P_{\text{MUSIC}}(\phi) = \frac{1}{\|E_n^H a(\phi)\|_2^2}, \tag{4}$$

where E_n is the estimated noise subspace and $a(\phi)$ is the steering vector for the parameter of interest. Peaks in this pseudospectrum indicate directions, ranges, or Doppler values consistent with the signal subspace.

The main appeal of MUSIC is that it can resolve targets more finely than FFT bin spacing when the covariance estimate is informative. The main risk is that covariance estimation degrades badly when sources are coherent or when the support is heavily masked.

FBSS addresses source coherence by averaging covariance matrices across overlapping subarrays and applying forward-backward symmetrization [5, 6]. In this project:

- spatial FBSS is always used in the active MUSIC path,
- range FBSS is used only when the known support contains a sufficiently long contiguous frequency run,
- and Doppler FBSS is used only when the known support contains a sufficiently long contiguous slow-time run.

This is an important point: the estimator is **support-aware**. It does not assume that every masked OFDM pattern admits a meaningful range or Doppler smoothing aperture.

3.4 Important Architectural Limitation

The active MUSIC method is staged rather than truly joint. It performs a dense azimuth search first, then conditioned range MUSIC, then conditioned Doppler MUSIC, and finally a local matched-filter refinement. It is therefore best described as a *staged masked MUSIC pipeline with FBSS*, not as a full joint 3-D MUSIC solver.

4 Study Design

4.1 Submission Bundle Used in This Report

All headline comparison claims in this report come from the saved submission bundle in the repository results tree. The report also uses saved follow-on 64-trial outputs for the expected-order and nuisance-strength diagnostics. Tables and figures are derived from CSV artifacts already written to disk.

The submission bundle used in the report has the following scope:

- FR1 anchor only,
- 64 trials per sweep point,
- 3 scene classes,
- 9 public sweep families,
- 42 sweep points per scene, plus a nominal point,
- per-trial CSV export for auditability.

The default final bundle is intentionally lean: it keeps the headline known-symbol comparison and removes pilot-only, representative-trial, and FBSS-ablation artifacts from the canonical submission path. Those diagnostics were useful during development, but they are not required for the final thesis claim set.

4.2 Waveform, Array, and Nominal Resource Mask

Table 1 summarizes the active waveform and array configuration for the submission bundle.

Table 1: Submission waveform and array configuration (FR1 anchor).

Parameter	Value
Carrier frequency	3.5 GHz
Occupied bandwidth	100 MHz
Subcarrier spacing	30 kHz
Physical OFDM subcarriers	3333
Simulated subcarriers used in the study	96
Snapshots in the balanced CPI	16
Slow-time snapshot spacing	0.5 ms (one 30 kHz slot)
Coherent processing interval	8 ms
Effective horizontal channels	16
Nominal range resolution	1.499 m
Nominal velocity resolution	5.353 m/s
Nominal azimuth resolution	6.769 deg
Nominal allocation family	Fragmented scheduled PRB
Nominal occupied fraction	0.500
Nominal contiguous bandwidth-span fraction	1.000
Nominal slow-time-span fraction	1.000
Nominal fragmentation index	0.0316

The nominal fragmented PRB grid is important. Even though only half of the resource elements are occupied, the occupied support spans the full simulated bandwidth and the full slow-time interval. In the implementation, the slow-time axis is sampled once per 30 kHz NR slot, so 16 snapshots correspond to an 8 ms CPI. This means the nominal point is not a "small bandwidth" or "short CPI" point; it is a *fragmented full-span* point.

4.3 Scenes and What Actually Separates Them

Table 2 summarizes the nominal scene settings. These scenes are not pure coherence cases. They differ in target pair, SNR, axis separations, target power imbalance, clutter count, multipath count, and configured target coherence.

Table 2: Nominal scene settings for the submission study.

Scene	Target pair	SNR (dB)	ΔR (m)	Δv (m/s)	$\Delta\theta$ (deg)	Power offset (dB)	Static / multipath
Open aisle	AMR + AMR	20	1.874	7.227	8.461	−1.0	2 / 1
Intersection	AMR + Forklift	18	1.499	6.692	10.491	+1.5	3 / 1
Rack aisle	AMR + AMR	16	1.274	4.818	6.092	−3.0	3 / 2

The configured target coherence values in the scenario generator are 0.15 for open aisle, 0.30 for intersection, and 0.65 for rack aisle. However, the saved trial logs show strongly overlapping *empirical* finite-snapshot coherence across scenes. In the submission bundle, the mean empirical target coherence is approximately 0.78 for open aisle, 0.77 for intersection, and 0.87 for rack aisle.

Figure 2 makes this overlap explicit by comparing the configured scene coherence with the empirical coherence realized in the nominal paired trial logs. Open aisle and intersection overlap heavily, and even rack aisle retains substantial overlap with both.

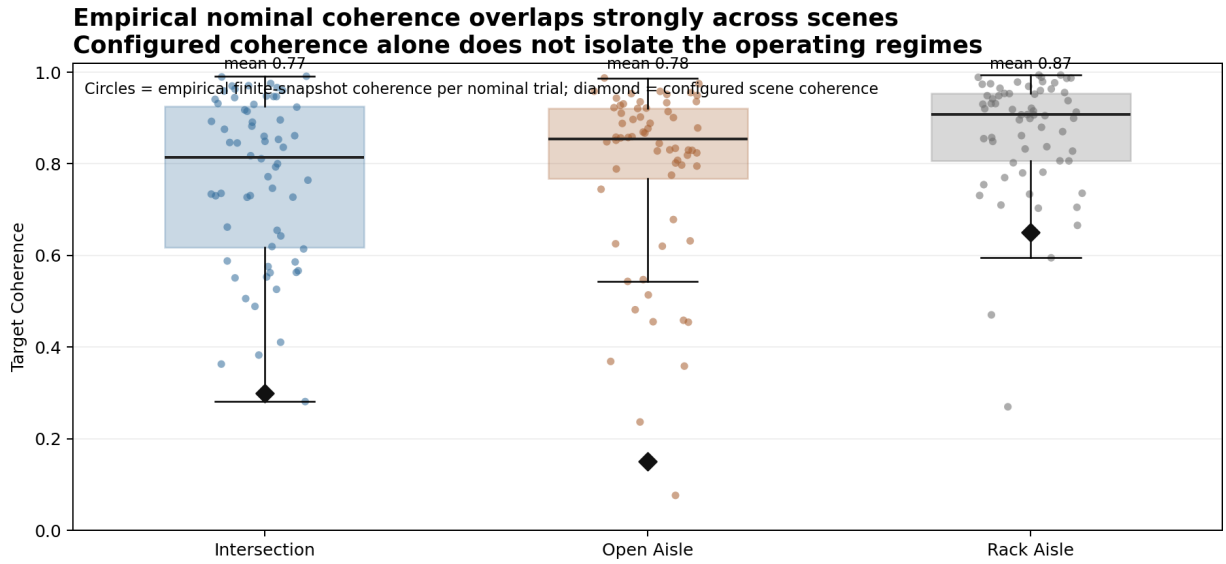

Figure 2: Configured versus empirical target coherence in the saved nominal paired trial logs. The empirical finite-snapshot coherence distributions overlap strongly across scenes, so the scene comparison should be interpreted as a composite-regime comparison rather than a clean coherence sweep.

This does **not** invalidate the measured estimator comparison. It does narrow the mechanism claim. The scenes should be interpreted as three composite operating regimes, not as a clean single-factor coherence sweep.

4.4 Public Sweep Families

The submission bundle evaluates nine public sweep families:

- allocation family,
- occupied fraction,
- fragmentation,
- bandwidth span,
- slow-time span,
- range separation,
- velocity separation,
- angle separation,
- nuisance gain offset (controlled clutter-strength study).

The separation sweeps are axis-isolated by design: when one axis is swept, the other two are held comfortably separated. This matters for interpretation later because a nominal scene win does not imply a win on every axis-isolated sweep.

5 Estimators and Evaluation Pipeline

5.1 Headline Comparison

The report centers on two headline methods:

1. `fft_masked`: a weighted masked FFT front end followed by local matched-filter refinement.
2. `music_masked`: a staged masked MUSIC estimator with spatial FBSS, support-aware range and Doppler FBSS when valid, and local matched-filter refinement.

This is a fair comparison in the sense that both methods use the same de-embedded masked observation, and the FFT baseline is not left as a raw coarse detector. The addition of local refinement to the FFT path is one of the reasons the final result is more balanced than the earlier optimistic picture.

5.2 Metrics

Each trial produces detections, assignments to truth targets, and summary metrics. The main probabilities are:

- P_{det}: both movers are detected within a detection gate,
- P_{joint}: both movers are assigned within the joint resolution gate,
- P_{range}, P_{vel}, and P_{angle}: analogous axis-specific resolution probabilities.

The trial evaluator uses normalized errors measured in range, velocity, and azimuth resolution cells. A joint-resolution success requires both movers to lie within the detection gate and within 0.35 of the nominal cell size on all three axes.

This 0.35-cell threshold is the active evaluation default from the current study configuration. Appendix B checks that the nominal headline verdict is stable under nearby thresholds using the saved per-trial assignments, without rerunning Monte Carlo.

The report also uses:

- false-alarm probability,
- miss probability,
- unconditional joint-assignment RMSE,
- paired nominal trial deltas between FFT and MUSIC.

For summary proportions, the bundle reports 95% Wilson confidence intervals. Because the same random seed spawn is used for both headline methods at a given sweep point, paired trial comparisons are meaningful.

6 Results

6.1 Nominal Submission Verdict

Table 3 summarizes the nominal FR1 submission point across the three scene classes.

Table 3: Nominal FR1 submission results, 64 trials per scene.

Scene	Method	P_{det}	P_{joint}	False alarm	Miss
Open aisle	FFT + local refinement	0.828	0.719	0.156	0.172
Open aisle	Staged MUSIC + FBSS	0.672	0.563	0.328	0.328
Intersection	FFT + local refinement	0.266	0.156	0.719	0.734
Intersection	Staged MUSIC + FBSS	0.844	0.703	0.156	0.156
Rack aisle	FFT + local refinement	0.047	0.031	0.938	0.953
Rack aisle	Staged MUSIC + FBSS	0.000	0.000	1.000	1.000

Figure 3 shows the same nominal result with Wilson intervals.

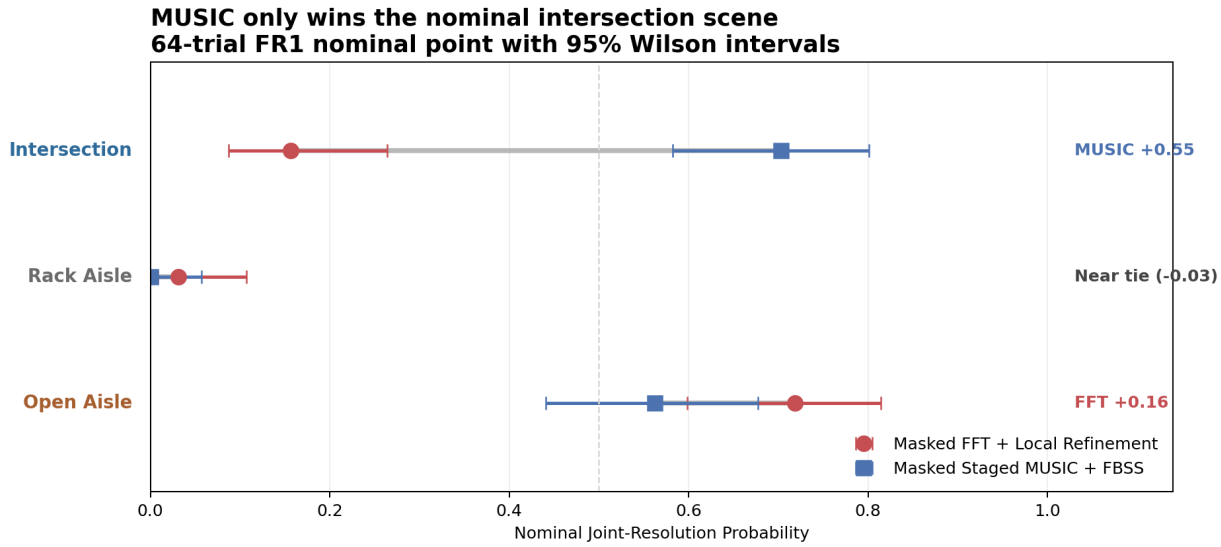

Figure 3: Nominal scene verdict from the saved submission bundle. The headline claim supported by the data is narrow: MUSIC wins only in the nominal intersection scene, loses in nominal open aisle, and does not produce a useful nominal result in rack aisle.

The nominal interpretation is straightforward:

- **Open aisle:** the fair FFT baseline outperforms MUSIC.
- **Intersection:** MUSIC outperforms FFT by a large margin.
- **Rack aisle:** both methods are poor, and MUSIC collapses completely at the nominal point.

This is already enough to reject any broad claim that MUSIC is generically better in the waveform-limited setting.

6.2 Paired Nominal Trial Evidence

The nominal averages are not being driven by one or two extreme outliers. Table 4 summarizes paired nominal trial outcomes from the saved trial logs.

Table 4: Paired nominal joint-resolution outcomes across 64 shared trials per scene.

Scene	FFT only	MUSIC only	Both succeed	Both fail
Open aisle	15	5	31	13
Intersection	0	35	10	19
Rack aisle	2	0	0	62

Figure 4 shows the same comparison in continuous form using paired unconditional joint-assignment RMSE deltas. Positive values favor MUSIC.

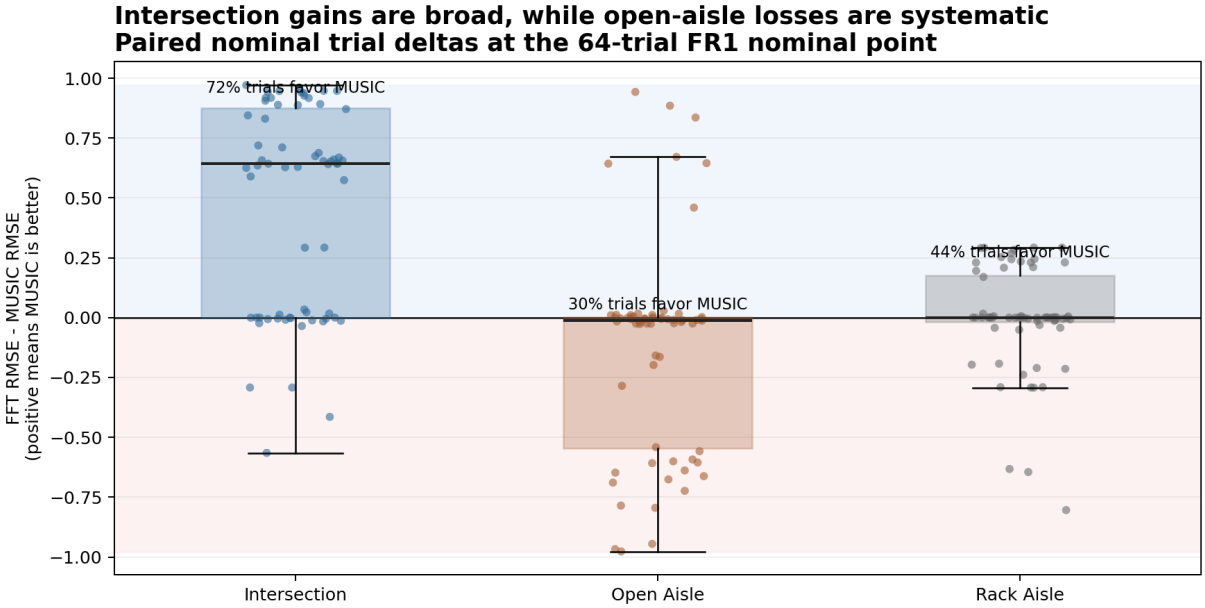

Figure 4: Paired nominal trial deltas from the saved submission bundle. The intersection advantage is broad, open-aisle losses are systematic, and rack aisle is mostly a shared failure region with a small number of mixed outcomes.

The paired-trial picture strengthens the nominal conclusion:

- in open aisle, only about 30% of trials favor MUSIC on the RMSE delta;
- in intersection, about 72% of trials favor MUSIC;
- in rack aisle, the comparison is dominated by shared failure rather than by a meaningful win region for either method.

6.3 Paired Statistical Support

Because the nominal headline methods share the same trial spawn keys, the submission bundle supports exact paired tests at the nominal point. Table 5 reports two such checks from the saved trial logs: an exact McNemar test on paired joint-resolution success and a two-sided exact sign test on paired unconditional joint-assignment RMSE deltas. Positive RMSE delta means MUSIC is better.

Table 5: Paired nominal significance checks from the saved 64-trial submission bundle.

Scene	McNemar p	RMSE sign-test p	Mean RMSE delta	Interpretation
Open aisle	0.041	1.6×10^{-3}	-0.124	FFT advantage supported
Intersection	5.8×10^{-11}	8.8×10^{-5}	$+0.459$	MUSIC advantage supported
Rack aisle	0.500	1.000	-0.005	No clear nominal winner

These tests do not change the story, but they make it more defensible. The nominal open-aisle loss and nominal intersection win are not just visually separated averages; they remain supported under paired nominal tests. Rack aisle does not support a stronger claim than "both methods mostly fail here."

6.4 Representative Negative Cases

The saved paired trial logs also support deterministic negative examples, which matter because they show that the project is not relying on a single favorable illustration. Table 6 summarizes two nominal paired trials taken directly from the saved `trial_level_results.csv` file.

Table 6: Representative negative nominal trials from the saved paired trial logs.

Scene	Spawn key	FFT outcome	MUSIC outcome
Open aisle	3	Both movers are assigned within the joint gate. The refined FFT detections remain close to both truth movers.	One mover is assigned correctly, but the second detection jumps to a distant spurious target at roughly 16 m and $-19.8°$, so joint success is lost.
Rack aisle	0	Joint failure. One detection is near the first mover, but the second is pulled far enough off the pair that both truth assignments fail the joint gate.	Joint failure. One mover is again recovered locally, while the second detection is diverted to a distant false target; this is a shared-failure clutter regime rather than a clean nominal win for either method.

These negative cases fit the aggregate picture. Open aisle contains real paired trials where FFT refinement is already sufficient and staged MUSIC overcommits to a spurious alternative, while rack aisle is dominated by hard cluttered trials in which neither method maintains two reliable joint assignments.

6.5 Regime Map Across Sweeps

The project becomes more interesting when it is interpreted as a regime map rather than as a single nominal verdict. Figure 5 summarizes the saved sweep families.

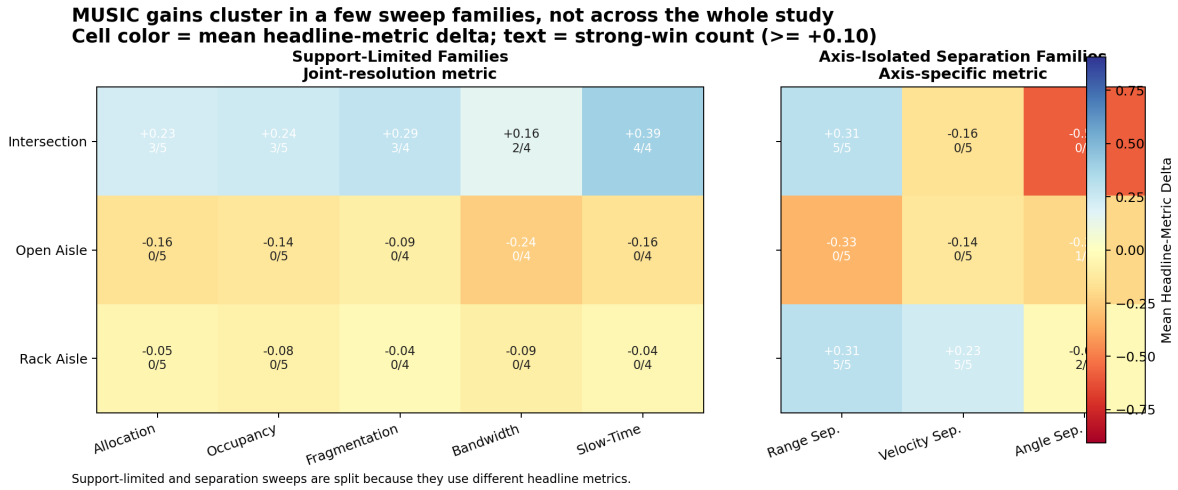

Figure 5: Two-panel regime map from the saved submission bundle. Support-limited sweeps (allocation, occupancy, fragmentation, bandwidth, slow time) are separated from axis-isolated separation sweeps because they use different headline metrics. Cell color shows the mean MUSIC-minus-FFT delta within that sweep family, while the text shows the number of strong-win points with delta at least +0.10.

Three sweep-level patterns matter:

1. **Open aisle:** MUSIC has no strong-win points at all in the 22 support-limited sweep points, and only one strong-win point in the 15 axis-isolated separation points. After FFT refinement, this is mostly an FFT-favorable regime.

2. **Intersection:** MUSIC wins broadly in the 22 support-limited sweep points, with 15 strong-win points and a positive mean delta in every support family. Its axis-specific advantage is much narrower: it helps on range separation, but not on the isolated velocity and angle sweeps.

3. **Rack aisle:** MUSIC does not help in the 22 support-limited sweep points and does not rescue the nominal point. Its positive cells are concentrated in easier separation sweeps, especially range and velocity separation, which means the staged pipeline can still benefit when the scene is made artificially easier.

The split regime map is the most important high-level result of the project. It says that MUSIC is neither universally useful nor universally useless under waveform limitation. Instead, its gains cluster in specific scene and support regimes, and the support-limited and axis-isolated sweep families must be interpreted separately.

6.6 Model-Order Diagnosis

The submission-bundle MDL estimates consistently overestimate the true source count of two. Mean estimated orders of 4–6 across scenes suggest that the masked covariance structure misleads the MDL criterion. To isolate this effect, the project includes an expected-order MUSIC variant that fixes $K = 2$.

Table 7 summarizes the saved 64-trial nominal comparison.

Table 7: Nominal P_{joint} comparison: MDL-estimated versus expected-order ($K = 2$) MUSIC.

Scene	MDL MUSIC	Expected-order MUSIC	FFT baseline
Open aisle	0.563	0.719	0.719
Intersection	0.703	0.844	0.156
Rack aisle	0.000	0.000	0.031

The result is decisive for two of three scenes. In open aisle, the original nominal loss to FFT ($P_{\text{joint}} = 0.563$ vs 0.719) was largely a model-order artifact: expected-order MUSIC recovers to exact parity with the FFT baseline. In intersection, expected-order MUSIC strengthens an already-clear win. In rack aisle, fixing $K = 2$ does not help at all, confirming that the failure mechanism is different.

6.7 Azimuth-Stage Failure Analysis

The saved stage diagnostics in the nominal rack-aisle bundle show why the scene resists the fix. In failed rack-aisle trials, the azimuth candidate list repeatedly contains a nuisance-aligned branch near $-21.7°$, close to the left-rack clutter at $-24°$, while the true second mover near $+5°$ is intermittently absent from the useful candidate set. The final detections then either duplicate the easier mover or jump to the nuisance branch.

Figure 6 visualises this directly. The left panel histograms the azimuth candidates produced across all 64 nominal rack-aisle trials. A persistent nuisance-aligned branch near $-21.7°$—close to the left-rack clutter at $-24°$—appears in every trial, while candidates near the two true movers ($\approx -1°$ and $+5°$) each appear in only half. The right panel shows where the final detections actually land: 50 of 128 detection slots (39%) are captured by the nuisance branch, while the true movers receive only 36 and 29 respectively.

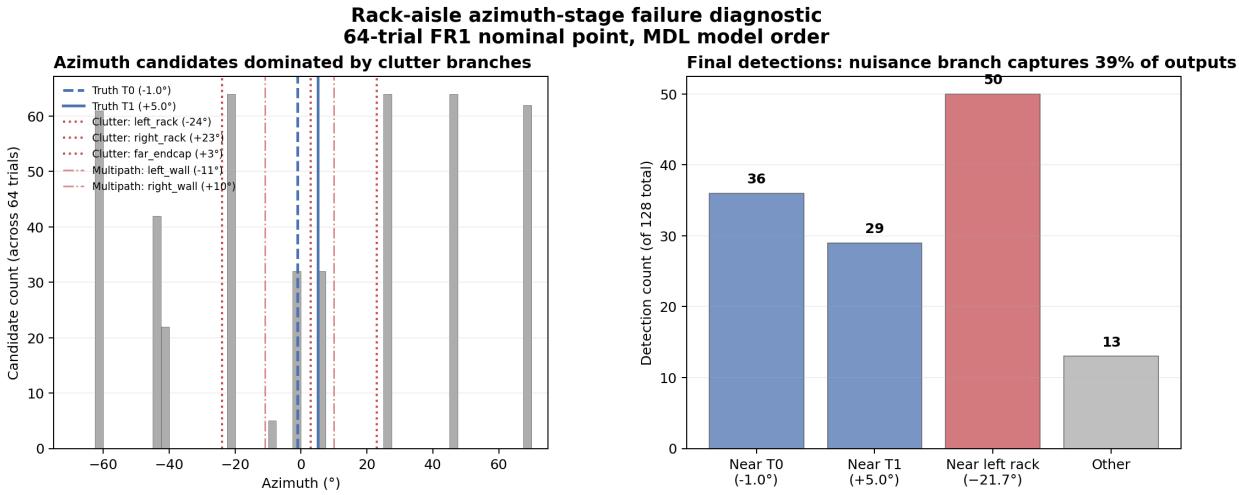

Figure 6: Rack-aisle azimuth-stage failure diagnostic. Left: histogram of azimuth candidates across 64 nominal trials, with truth target azimuths (blue dashed/solid) and clutter/multipath angles (red dotted/dash-dot). The persistent candidate near $-21.7°$ aligns with the left-rack clutter at $-24°$ and appears in every trial. Right: distribution of final detection azimuths showing that the nuisance branch captures 39% of outputs, confirming spatial subspace capture as the failure mechanism.

Combined with the expected-order result staying at $P_{\text{joint}} = 0$, this indicates that rack aisle is not primarily a model-order problem. The more plausible interpretation is a clutter-dominated azimuth-stage failure: nuisance scatterers capture the spatial subspace early enough that later range/Doppler processing cannot recover the missing second mover.

6.8 Controlled Nuisance-Strength Study

To test the clutter explanation more directly, the project includes a nuisance-gain-offset sweep that shifts all nuisance scatterer gains uniformly while holding the scene geometry, target placement, and nominal separations fixed. The saved sweep evaluates offsets of −6, −3, 0, +3, and +6 dB relative to the scene defaults.

The cleaner version of this test is the expected-order sweep, because it removes the MDL confound. Table 8 summarizes the saved expected-order P_{joint} values.

Table 8: Expected-order nuisance sweep: joint-resolution probability P_{joint} under uniform nuisance-gain offsets.

Scene	Method	−6 dB	−3 dB	0 dB	+3 dB	+6 dB
Open aisle	FFT + local refinement	0.719	0.766	0.750	0.688	0.672
Open aisle	Expected-order MUSIC	0.734	0.703	0.766	0.484	0.406
Intersection	FFT + local refinement	0.125	0.234	0.172	0.219	0.125
Intersection	Expected-order MUSIC	0.828	0.781	0.797	0.797	0.781
Rack aisle	FFT + local refinement	0.047	0.047	0.031	0.000	0.000
Rack aisle	Expected-order MUSIC	0.000	0.000	0.000	0.000	0.000

This produces the clearest regime-boundary evidence in the project:

- **Open aisle:** once model order is fixed, MUSIC is competitive or slightly better than FFT at low-to-moderate nuisance levels, but it degrades sharply when clutter is amplified.
- **Intersection:** expected-order MUSIC remains strong across the entire tested nuisance range and keeps a large margin over FFT even at +6 dB.
- **Rack aisle:** MUSIC remains at zero throughout, which supports the interpretation that this scene is already beyond the clutter tolerance of the staged pipeline.

The corresponding MDL nuisance sweep is useful mainly as a contrast case. In intersection, the MDL path deteriorates sharply as nuisance strength increases and falls to $P_{\text{joint}} = 0.031$ at +6 dB, whereas the expected-order path stays strong at $P_{\text{joint}} = 0.781$ at the same point. Rack aisle still shows no rescue under either model-order mode. The expected-order table above is therefore the more defensible causal view because it isolates clutter effects from model-order error.

7 Discussion

7.1 What the Project Actually Supports

The strongest defensible claim is:

> MUSIC has real super-resolution capability in realistic OFDM ISAC, but its value is regime-dependent. Under the default MDL path it wins strongly only in intersection;

> with correct model order it also becomes competitive in open aisle; and in clutter-dominated rack aisle it fails regardless of model order.

This is a stronger and more useful result than a simple "MUSIC wins" or "MUSIC loses" claim. It identifies *why* the super-resolution upside seen in the earlier 1-D range-only study does not always survive into realistic 3-D operation, and it points to specific adaptations and limits: model-order control can recover part of the lost capability, while nuisance-dominated spatial subspaces define a harder regime boundary.

7.2 What Needs Honest Framing

Two framing points matter.

First, the scenes should not be described as if target coherence were the only or even the dominant controlled difference. They are composite regimes that differ in SNR, nominal separations, target pair, target power imbalance, clutter count, and multipath layout. Because the saved empirical coherence overlaps strongly across scenes in Figure 2, the report avoids attributing the scene results to coherence alone.

Second, expected-order MUSIC should be framed correctly. It is an oracle diagnostic for this fixed two-target study, not a deployable model-order solution. Its value is that it reveals what performance is available once subspace dimensioning is no longer the dominant failure mode.

These points narrow the mechanism claim, but they do **not** invalidate the empirical comparison. The saved CSVs still support the estimator comparison reported here.

7.3 Applicability to 6G

The result is useful to a 6G-oriented audience for three reasons:

1. It studies sensing from the communication waveform itself rather than from a radar-only waveform.
2. It uses support limitations that look like scheduling decisions rather than ideal contiguous radar occupancy.
3. It gives a regime map, which is more actionable for system design than a single broad positive claim.

The practical message is not "deploy MUSIC everywhere." It is closer to:

> MUSIC is a conditional tool for communications-limited sensing. It is worth considering where scene geometry, clutter strength, and support conditions create both a genuine super-resolution need and a usable covariance structure. It is not a default replacement for a strong FFT baseline.

The saved runtime summary also sharpens the engineering interpretation. At the nominal FR1 point, staged MUSIC adds about 0.19–0.26 s of total runtime per sweep point relative to FFT, raising total point runtime from roughly 0.76 s to 0.96–0.99 s, or about 25–30%. Because both methods share the same masked front end and much of the preprocessing cost, the main objection to MUSIC in this study is still not runtime alone. The more important question is whether a given operating regime justifies that extra cost.

7.4 Relation to Fundamental Bounds

This project does not compute the Cramér-Rao bound (CRB) for the masked observation model. The standard CRB for MUSIC-based direction-of-arrival estimation under known model order is well-established [7], and MUSIC is known to approach CRB efficiency asymptotically in the uncorrelated, full-support case. However, two features of this problem complicate a direct CRB computation:

First, the fragmented support mask $m_{k,n}$ creates a non-standard Fisher information structure. The observation likelihood depends on which resource elements are known, and the effective Fisher information matrix is a function of the support pattern, not just the array geometry and SNR. Computing the CRB for each sweep point would require evaluating the Fisher information under the specific mask realization, which is tractable but beyond the scope of this project.

Second, the staged architecture decomposes the joint 3-D estimation problem into sequential 1-D problems. The CRB for the staged estimator is not the same as the CRB for the joint problem, and the gap between them depends on the conditioning error propagated from earlier stages. This staging loss is itself a quantity of interest that the project measures empirically but does not bound analytically.

For context, the unconditional joint-assignment RMSE values observed in the submission bundle (0.2–0.5 normalized resolution cells at the nominal point) are well above what classical CRB analysis would predict for a full-support, known-order scenario at comparable SNR. This gap is consistent with the project's main finding: waveform limitation and model-order misestimation are the dominant performance limiters, not fundamental information bounds.

7.5 Important Limitations

Several limitations remain and should be stated plainly:

- The active estimator is staged, not true joint 3-D MUSIC.
- The study is limited to two movers and a small number of scene templates.
- The scene classes are composite regimes rather than a clean one-factor coherence sweep.
- The bundle is FR1-only for the final submission snapshot.
- MDL-based model-order estimation fails systematically under masked covariance, consistently overestimating the source count. The expected-order variant shows what is recoverable when that error is removed, but a practical system would still need a robust masked-covariance model-order method.
- The study is simulation-only and does not include hardware, channel estimation error, or communication-link feedback constraints.

8 Conclusion

This project set out to determine whether the super-resolution upside of MUSIC, previously demonstrated in a 1-D range-only study, survives into waveform-limited 3-D OFDM ISAC operation with fragmented waveforms, structured clutter, and coherent multipath.

The answer is conditional but well-characterized:

- Under the default MDL path, MUSIC retains genuine super-resolution capability only in favorable regimes, decisively outperforming the FFT baseline in intersection while losing in open aisle and collapsing in rack aisle.
- The original nominal open-aisle loss was substantially a model-order artifact. Expected-order MUSIC recovers to parity with FFT there and strengthens the already-clear intersection win, demonstrating that the underlying super-resolution machinery is still present once subspace dimensioning is controlled.
- Rack aisle represents a structurally different failure: nuisance-aligned azimuth candidates dominate the staged search, and no model-order fix recovers performance.
- The expected-order nuisance sweep reveals three distinct regimes: open aisle is conditionally favorable only before clutter becomes too strong, intersection remains robust across the tested nuisance range, and rack aisle stays beyond the useful operating boundary throughout.

The central contribution is therefore a resolution-versus-robustness map for MUSIC in realistic OFDM ISAC. The project identifies two key failure mechanisms (model-order overestimation under masked covariance, and nuisance-dominated spatial subspace capture) and shows that the first is correctable while the second defines a hard regime boundary.

Future work should address:

1. robust model-order estimation methods adapted to masked OFDM covariance (e.g., information-theoretic criteria with penalty correction for missing support),
2. clutter-aware subspace methods that can separate nuisance contributions before the staged search,
3. and a receiver path for limited symbol knowledge beyond zero-filling.

Within its scope, the project provides an honest and mechanistically grounded study of where and why subspace super-resolution helps in communications-limited OFDM ISAC.

A Estimator And Evaluation Settings

Table 9 collects the most important implementation settings for the active headline comparison. The purpose is not to restate every runtime parameter, but to make the fairness assumptions and evaluation defaults explicit in one place.

Table 9: Key implementation settings for the active headline comparison.

Setting	Active value / interpretation
Expected truth target count	2 movers per trial
FFT front end	Weighted masked FFT on the same de-embedded known-symbol cube used by MUSIC
FFT candidate extraction	Noise-floor thresholding with scale 7.5 and candidate backfill pool size 128
Detection non-maximum suppression radius	0.9 resolution cells
Headline FFT back end	Local matched-filter refinement on coarse FFT candidates
Headline MUSIC architecture	Staged dense azimuth → conditioned range → conditioned Doppler search, followed by local matched-filter refinement
Spatial FBSS	Always enabled in the active MUSIC path
Range / Doppler FBSS	Enabled only when the support pattern admits a valid smoothing aperture
Model-order handling	MDL-estimated (default) or expected-order ($K = 2$) mode; MDL estimate is logged and used to widen the azimuth hypothesis pool; final reported target count remains capped by the fixed two-mover study setup
Paired comparison basis	Shared trial spawn keys across headline methods at each sweep point
Joint-resolution gate	Detection gate on all three axes, then 0.35 resolution-cell threshold on each axis

B Nominal Gate Sensitivity

Table 10 recomputes nominal joint-resolution probabilities from the saved per-trial assignment logs using several nearby joint-resolution thresholds. No Monte Carlo trials are rerun. The key point is that the nominal verdict is stable: open aisle continues to favor FFT, intersection continues to favor MUSIC, and rack aisle remains a shared-failure region throughout this threshold range.

Table 10: Nominal joint-resolution probability under several joint gate thresholds, recomputed from saved trial assignments only.

Scene	Method	0.25 cell	0.35 cell	0.45 cell	0.50 cell
Open aisle	FFT + local refinement	0.656	0.719	0.766	0.781
Open aisle	Staged MUSIC + FBSS	0.531	0.562	0.578	0.578
Intersection	FFT + local refinement	0.125	0.156	0.219	0.219
Intersection	Staged MUSIC + FBSS	0.641	0.703	0.750	0.781
Rack aisle	FFT + local refinement	0.031	0.031	0.031	0.047
Rack aisle	Staged MUSIC + FBSS	0.000	0.000	0.000	0.000

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